

Nonlinear Simplicial Waves and Anomalous Porous Transport Induced by Beta-Kernel Fractional Laplacians

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Abstract

We develop a comprehensive analytical and mathematical physics framework for non-local wave dynamics and subdiffusive transport governed by the *Fractional Simplicial Laplacian* $\Delta_{\Delta_m}^\alpha$, an operator induced by continuous multinomial Beta-kernels on the standard simplex $\Delta_{m-1}(\alpha)$. In the first part of this work, we formulate the *Nonlinear Simplicial Schrödinger Equation* (NLSE). We prove the global conservation of particle mass $\mathcal{N}[\psi]$ and simplicial Hamiltonian energy $\mathcal{E}[\psi]$ via the self-adjointness of the continuous Beta-integral kernel. We perform a modulational instability analysis demonstrating that plane waves undergo directional sideband destabilization governed by the discrete S_m point-group symmetry of the simplex, producing facet-aligned anisotropic solitary waves. In the second part, we formulate the space-time fractional simplicial diffusion equation describing anomalous transport in anisotropic porous, fractured, and crystalline media. Using Fourier–Laplace transform methods, we derive the exact non-local propagator in terms of Mittag-Leffler functions and rigorously prove that the long-time Mean Squared Displacement (MSD) covariance tensor scales as $\langle \mathbf{x}\mathbf{x}^T \rangle(t) = \frac{\mathcal{K}_{\text{diff}}}{m\alpha\Gamma(\beta+1)} \mathbf{A}_{m-1} t^\beta$, where $\mathbf{A}_{m-1}$ is the Cartan matrix of the Lie algebra A_{m-1} . This establishes a rigorous geometric connection between Lie algebraic root systems, compact-support memory kernels, and anisotropic anomalous diffusion in structured media.

Contents

1	Introduction and Physical Motivation	2
2	The Fractional Simplicial Laplacian and its Metric Geometry	2
3	Nonlinear Simplicial Schrödinger Equations (NLSE)	3
3.1	Fundamental Conservation Laws	3
3.2	Modulational Instability on Simplicial Torus	4
3.3	Simplicial Solitary Waves and Anisotropic Solitons	4
4	Anomalous Diffusion in Anisotropic Porous Media	5
4.1	Model Formulation	5
4.2	Exact Propagator via Mittag-Leffler Functions	5
4.3	Mean Squared Displacement (MSD) and Cartan Permeability Tensor	5
5	Conclusion	6

1 Introduction and Physical Motivation

Non-local partial differential equations (PDEs) and fractional calculus have become foundational tools for describing memory-dependent dynamics, anomalous transport, and non-local wave phenomena across quantum physics, fluid mechanics, and geophysical transport [2–5]. In traditional fractional quantum mechanics and anomalous diffusion models, the non-local spatial operator is typically chosen as the isotropic Riesz fractional Laplacian $(-\Delta)^{\alpha/2}$, whose integral representation:

$$(-\Delta)^{\alpha/2}u(\mathbf{x}) = C_{d,\alpha} \text{P.V.} \int_{\mathbb{R}^d} \frac{u(\mathbf{x}) - u(\mathbf{y})}{|\mathbf{x} - \mathbf{y}|^{d+\alpha}} d\mathbf{y}, \quad (1.1)$$

is isotropic, radially symmetric under the orthogonal group $O(d)$, and characterized by algebraic power-law tails that extend across the entire space $\mathbb{R}^d$.

However, in many realistic physical media—such as anisotropic fractured rocks, crystalline waveguides, packed colloidal arrays, and simplicial metamaterials—the underlying microstructure imposes two fundamental physical constraints:

1. **Directional Geometric Anisotropy:** Transport occurs predominantly along structured interconnected channels with characteristic intersection angles (such as triangular or tetrahedral lattices), breaking continuous rotational symmetry.
2. **Finite Horizon of Non-Local Steps:** Microscopic jump lengths per collision event are geometrically bounded by the pore or cell size, precluding unphysical infinite-velocity jump tails.

To bridge this fundamental gap, we introduce and analyze dynamical and transport processes governed by the *Fractional Simplicial Laplacian* Δ_m^α , constructed from the continuous multinomial Beta-kernel on the standard $(m-1)$ -simplex $\Delta_{m-1}(\alpha)$.

2 The Fractional Simplicial Laplacian and its Metric Geometry

Let $\mathbf{x} = (x_1, \dots, x_{m-1}) \in \mathbb{R}^{m-1}$ denote spatial coordinates, and let $\Delta_{m-1}(\alpha) \subset \mathbb{R}_{\geq 0}^{m-1}$ be the standard scaled simplex of size $\alpha > 0$:

$$\Delta_{m-1}(\alpha) := \left\{ \mathbf{y} \in \mathbb{R}_{\geq 0}^{m-1} \left| \sum_{j=1}^{m-1} y_j \leq \alpha \right. \right\}. \quad (2.1)$$

The dependent barycentric coordinate is defined by $y_m := \alpha - \sum_{j=1}^{m-1} y_j$. The continuous multinomial Beta-kernel is defined by

$$\binom{\alpha}{\mathbf{y}} := \frac{\Gamma(\alpha+1)}{\prod_{j=1}^m \Gamma(y_j+1)}, \quad I_m(\alpha) := \int_{\Delta_{m-1}(\alpha)} \binom{\alpha}{\mathbf{y}} d\mathbf{y}. \quad (2.2)$$

Definition 2.1 (Fractional Simplicial Laplacian). For fractional parameter $\alpha > 0$, the self-adjoint Fractional Simplicial Laplacian acting on $u \in H^2(\mathbb{R}^{m-1})$ is defined by

$$\Delta_{\Delta_m}^\alpha u(\mathbf{x}) := \frac{1}{2\alpha^2 I_m(\alpha)} \int_{\Delta_{m-1}(\alpha)} \binom{\alpha}{\mathbf{y}} \left[u(\mathbf{x} + \mathbf{y}) - 2u(\mathbf{x}) + u(\mathbf{x} - \mathbf{y}) \right] d\mathbf{y}. \quad (2.3)$$

Proposition 2.2 (Self-Adjointness and Positivity). *The operator $-\Delta_{\Delta_m}^\alpha$ is linear, symmetric, and positive semi-definite on $L^2(\mathbb{R}^{m-1})$:*

$$\langle u, -\Delta_{\Delta_m}^\alpha u \rangle_{L^2} = \frac{1}{2\alpha^2 I_m(\alpha)} \int_{\mathbb{R}^{m-1}} d\mathbf{x} \int_{\Delta_{m-1}(\alpha)} \binom{\alpha}{\mathbf{y}} |u(\mathbf{x} + \mathbf{y}) - u(\mathbf{x})|^2 d\mathbf{y} \geq 0. \quad (2.4)$$

Proof. Expanding the inner product $\langle u, -\Delta_{\Delta_m}^\alpha u \rangle_{L^2} = \int_{\mathbb{R}^{m-1}} u(\mathbf{x})(-\Delta_{\Delta_m}^\alpha u(\mathbf{x}))d\mathbf{x}$:

$$\langle u, -\Delta_{\Delta_m}^\alpha u \rangle = \frac{1}{2\alpha^2 I_m(\alpha)} \int_{\Delta_{m-1}(\alpha)} \binom{\alpha}{\mathbf{y}} \left[\int_{\mathbb{R}^{m-1}} \left(2|u(\mathbf{x})|^2 - u(\mathbf{x})u(\mathbf{x} + \mathbf{y}) - u(\mathbf{x})u(\mathbf{x} - \mathbf{y}) \right) d\mathbf{x} \right] d\mathbf{y}.$$

Using the translation invariance $\int u(\mathbf{x})u(\mathbf{x} \pm \mathbf{y})d\mathbf{x} = \int u(\mathbf{x} \mp \mathbf{y})u(\mathbf{x})d\mathbf{x}$:

$$\int_{\mathbb{R}^{m-1}} \left(2|u(\mathbf{x})|^2 - 2u(\mathbf{x})u(\mathbf{x} + \mathbf{y}) \right) d\mathbf{x} = \int_{\mathbb{R}^{m-1}} |u(\mathbf{x} + \mathbf{y}) - u(\mathbf{x})|^2 d\mathbf{x}. \quad (2.5)$$

Since $\binom{\alpha}{\mathbf{y}} > 0$ on the interior of $\Delta_{m-1}(\alpha)$, the integral is strictly non-negative, vanishing if and only if u is constant. $\square$

Theorem 2.3 (Dispersion Symbol and Cartan Metric Emergence). *Let $\mathbf{k} = (k_1, \dots, k_{m-1}) \in \mathbb{R}^{m-1}$ with the affine simplicial projection $k_m \equiv -\sum_{j=1}^{m-1} k_j$ (enforcing $\sum_{j=1}^m k_j = 0$ on the simplex root hyperplane). The Fourier symbol $\sigma_{\Delta_m}^\alpha(\mathbf{k})$ defined by $-\Delta_{\Delta_m}^\alpha e^{i\mathbf{k} \cdot \mathbf{x}} = \sigma_{\Delta_m}^\alpha(\mathbf{k})e^{i\mathbf{k} \cdot \mathbf{x}}$ is given explicitly by*

$$\sigma_{\Delta_m}^\alpha(\mathbf{k}) = \frac{1}{\alpha^2} [1 - R(\mathbf{k})^\alpha \cos(\alpha\Theta(\mathbf{k}))], \quad (2.6)$$

where $R(\mathbf{k}) := \frac{1}{m} \left| \sum_{j=1}^m e^{-ik_j} \right| \leq 1$ and $\Theta(\mathbf{k}) := \text{Arg} \left(\sum_{j=1}^m e^{-ik_j} \right)$.

In the long-wavelength limit $|\mathbf{k}| \rightarrow 0$:

$$\lim_{|\mathbf{k}| \rightarrow 0} \sigma_{\Delta_m}^\alpha(\mathbf{k}) = \frac{1}{2m\alpha} \mathbf{k}^T \mathbf{A}_{m-1} \mathbf{k}, \quad (2.7)$$

where $\mathbf{A}_{m-1}$ is the $(m-1) \times (m-1)$ Cartan matrix of the simple Lie algebra A_{m-1} :

$$\mathbf{A}_{m-1} = \begin{pmatrix} 2 & 1 & 1 & \cdots & 1 \\ 1 & 2 & 1 & \cdots & 1 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & 1 & 1 & \cdots & 2 \end{pmatrix}. \quad (2.8)$$

3 Nonlinear Simplicial Schrödinger Equations (NLSE)

We formulate the Cauchy problem for the non-local Fractional Simplicial NLSE on $\mathbb{R}^{m-1} \times \mathbb{R}^+$:

$$i\hbar \frac{\partial \psi(\mathbf{x}, t)}{\partial t} = -\frac{\hbar^2}{2M} \Delta_{\Delta_m}^\alpha \psi(\mathbf{x}, t) + V(\mathbf{x})\psi(\mathbf{x}, t) - \kappa |\psi(\mathbf{x}, t)|^{2\sigma} \psi(\mathbf{x}, t), \quad (3.1)$$

where $M > 0$ is the effective mass, $V(\mathbf{x})$ is an external real-valued potential, $\kappa \in \mathbb{R}$ is the non-linear coupling constant ($\kappa > 0$ focusing, $\kappa < 0$ defocusing), and $\sigma > 0$ is the non-linearity exponent.

3.1 Fundamental Conservation Laws

Theorem 3.1 (Global Conservation of Mass and Energy). *Let $\psi(\mathbf{x}, t)$ be a smooth solution to (3.1) decaying sufficiently fast at spatial infinity. Then:*

1. **Conservation of Total Mass (Particle Number):**

$$\mathcal{N}[\psi(t)] := \int_{\mathbb{R}^{m-1}} |\psi(\mathbf{x}, t)|^2 d\mathbf{x} = \mathcal{N}[\psi(0)]. \quad (3.2)$$

2. **Conservation of Simplicial Energy (Hamiltonian):**

$$\begin{aligned} \mathcal{E}[\psi(t)] &:= \frac{\hbar^2}{2M} \int_{\mathbb{R}^{m-1}} \sigma_{\Delta_m}^\alpha(\mathbf{k}) |\hat{\psi}(\mathbf{k}, t)|^2 d\mathbf{k} + \int_{\mathbb{R}^{m-1}} V(\mathbf{x}) |\psi(\mathbf{x}, t)|^2 d\mathbf{x} \\ &\quad - \frac{\kappa}{\sigma + 1} \int_{\mathbb{R}^{m-1}} |\psi(\mathbf{x}, t)|^{2\sigma+2} d\mathbf{x} = \mathcal{E}[\psi(0)]. \end{aligned} \quad (3.3)$$

Proof.

1. Multiplying (3.1) by ψ^* and taking the imaginary part:

$$\frac{\hbar}{2} \frac{\partial |\psi|^2}{\partial t} = \text{Im} \left(-\frac{\hbar^2}{2M} \psi^* \Delta_{\Delta_m}^\alpha \psi \right). \quad (3.4)$$

Integrating over $\mathbb{R}^{m-1}$ and applying the self-adjointness of $\Delta_{\Delta_m}^\alpha$:

$$\frac{d}{dt} \mathcal{N}[\psi(t)] = -\frac{\hbar}{M} \text{Im} \langle \psi, \Delta_{\Delta_m}^\alpha \psi \rangle_{L^2} = 0, \quad (3.5)$$

since $\langle \psi, \Delta_{\Delta_m}^\alpha \psi \rangle_{L^2} \in \mathbb{R}$.

2. Differentiating $\mathcal{E}[\psi(t)]$ with respect to t :

$$\frac{d\mathcal{E}}{dt} = 2 \text{Re} \int_{\mathbb{R}^{m-1}} \psi_t^* \left[-\frac{\hbar^2}{2M} \Delta_{\Delta_m}^\alpha \psi + V(\mathbf{x})\psi - \kappa |\psi|^{2\sigma} \psi \right] d\mathbf{x}. \quad (3.6)$$

Substituting the equation of motion $i\hbar\psi_t = \mathcal{H}\psi$:

$$\frac{d\mathcal{E}}{dt} = 2 \text{Re} \int_{\mathbb{R}^{m-1}} \left(\frac{1}{i\hbar} \mathcal{H}\psi \right)^* (\mathcal{H}\psi) d\mathbf{x} = 2 \text{Re} \left(-\frac{1}{i\hbar} \|\mathcal{H}\psi\|_{L^2}^2 \right) = 0. \quad (3.7)$$

□

3.2 Modulational Instability on Simplicial Torus

Consider a uniform plane-wave background state with amplitude $\psi_0(\mathbf{x}, t) = \sqrt{\rho_0} e^{-i\mu_0 t/\hbar}$, where the chemical potential is $\mu_0 = -\kappa\rho_0^\sigma$. We perturb the background as

$$\psi(\mathbf{x}, t) = (\sqrt{\rho_0} + \delta u(\mathbf{x}, t) + i\delta v(\mathbf{x}, t)) e^{-i\mu_0 t/\hbar}. \quad (3.8)$$

Theorem 3.2 (Simplicial Modulational Instability). *Linearizing (3.1) with respect to perturbations $(\delta u, \delta v) \propto e^{i(\mathbf{k} \cdot \mathbf{x} - \Omega t)}$, the perturbation frequency satisfies the dispersion relation:*

$$\hbar^2 \Omega^2(\mathbf{k}) = \frac{\hbar^2 \sigma_{\Delta_m}^\alpha(\mathbf{k})}{2M} \left[\frac{\hbar^2 \sigma_{\Delta_m}^\alpha(\mathbf{k})}{2M} - 2\kappa\sigma\rho_0^\sigma \right]. \quad (3.9)$$

In the focusing regime ($\kappa > 0$), the plane wave is modulationally unstable ($\Omega^2 < 0$) for wavevectors satisfying

$$\sigma_{\Delta_m}^\alpha(\mathbf{k}) < \frac{4M\kappa\sigma\rho_0^\sigma}{\hbar^2}, \quad (3.10)$$

with maximum instability growth rate $\gamma_{\max} = \frac{\kappa\sigma\rho_0^\sigma}{\hbar}$ attained on the simplicial resonant manifold.

3.3 Simplicial Solitary Waves and Anisotropic Solitons

For stationary solitary wave states $\psi(\mathbf{x}, t) = e^{-i\mu t/\hbar} \phi(\mathbf{x})$ with $\mu < 0$:

$$-\frac{\hbar^2}{2M} \Delta_{\Delta_m}^\alpha \phi(\mathbf{x}) + (V(\mathbf{x}) - \mu)\phi(\mathbf{x}) = \kappa |\phi(\mathbf{x})|^{2\sigma} \phi(\mathbf{x}). \quad (3.11)$$

Theorem 3.3 (Symmetry of Simplicial Solitons). *Unlike standard fractional solitons that are radially symmetric under $O(m-1)$, the ground state solutions $\phi(\mathbf{x})$ of (3.11) are invariant under the discrete permutation and reflection group of the simplex S_m . Along the principal axes $\mathbf{e}_j - \mathbf{e}_k$, the decay profile exhibits facet-aligned algebraic directional decay rates.*

4 Anomalous Diffusion in Anisotropic Porous Media

4.1 Model Formulation

In heterogeneous porous and fractured geological media, microscopic transport is characterized by: (i) temporal retention in pore dead-ends (subdiffusion), and (ii) anisotropic spatial hopping constrained by simplicial pore geometries.

Definition 4.1 (Space-Time Fractional Simplicial Diffusion Equation). Let $u(\mathbf{x}, t)$ denote the solute concentration. The governing equation is

$$\frac{\partial^\beta u(\mathbf{x}, t)}{\partial t^\beta} = \mathcal{K}_{\text{diff}} \Delta_{\Delta_m}^\alpha u(\mathbf{x}, t), \quad \mathbf{x} \in \mathbb{R}^{m-1}, \quad t > 0, \quad (4.1)$$

with initial condition $u(\mathbf{x}, 0) = u_0(\mathbf{x})$, where $\beta \in (0, 1]$ is the temporal order of the Caputo fractional derivative:

$$\frac{\partial^\beta u}{\partial t^\beta}(t) := \frac{1}{\Gamma(1-\beta)} \int_0^t (t-\tau)^{-\beta} \frac{\partial u}{\partial \tau}(\tau) d\tau. \quad (4.2)$$

4.2 Exact Propagator via Mittag-Leffler Functions

Theorem 4.2 (Closed-Form Solution in Frequency Domain). *The spatial Fourier transform of the exact solution to (4.1) is given by*

$$\widehat{u}(\mathbf{k}, t) = E_\beta \left(-\mathcal{K}_{\text{diff}} \sigma_{\Delta_m}^\alpha(\mathbf{k}) t^\beta \right) \widehat{u}_0(\mathbf{k}), \quad (4.3)$$

where $E_\beta(z) := \sum_{n=0}^{\infty} \frac{z^n}{\Gamma(\beta n + 1)}$ is the Mittag-Leffler function.

Proof. Applying the Laplace transform $\mathcal{L}\{f\}(s) = \int_0^\infty e^{-st} f(t) dt$ to the Caputo derivative yields $\mathcal{L}\{\partial_t^\beta u\}(s) = s^\beta \widehat{u}(s) - s^{\beta-1} u(0)$. Applying the spatial Fourier transform $\mathbf{x} \rightarrow \mathbf{k}$:

$$s^\beta \widehat{u}(\mathbf{k}, s) - s^{\beta-1} \widehat{u}_0(\mathbf{k}) = -\mathcal{K}_{\text{diff}} \sigma_{\Delta_m}^\alpha(\mathbf{k}) \widehat{u}(\mathbf{k}, s). \quad (4.4)$$

Solving algebraically for $\widehat{u}(\mathbf{k}, s)$:

$$\widehat{u}(\mathbf{k}, s) = \frac{s^{\beta-1}}{s^\beta + \mathcal{K}_{\text{diff}} \sigma_{\Delta_m}^\alpha(\mathbf{k})} \widehat{u}_0(\mathbf{k}). \quad (4.5)$$

Applying the inverse Laplace transform using the standard identity $\mathcal{L}^{-1} \left\{ \frac{s^{\beta-1}}{s^\beta + a} \right\}(t) = E_\beta(-at^\beta)$ immediately yields the result. $\square$

4.3 Mean Squared Displacement (MSD) and Cartan Permeability Tensor

Theorem 4.3 (Anisotropic Mean Squared Displacement). *Assume a localized initial condition $u_0(\mathbf{x}) = \delta(\mathbf{x})$. The first and second moments of the concentration profile satisfy:*

1. **Mean Drift:** $\langle \mathbf{x}(t) \rangle = \mathbf{0}$.

2. **Mean Squared Displacement Covariance Tensor:**

$$\langle \mathbf{x} \mathbf{x}^T \rangle(t) := \int_{\mathbb{R}^{m-1}} \mathbf{x} \mathbf{x}^T u(\mathbf{x}, t) d\mathbf{x} = \frac{\mathcal{K}_{\text{diff}}}{m\alpha \Gamma(\beta+1)} \mathbf{A}_{m-1} \cdot t^\beta. \quad (4.6)$$

Proof. Using moment-generating properties of the Fourier transform:

$$\langle x_j \rangle(t) = i \frac{\partial \widehat{u}(\mathbf{k}, t)}{\partial k_j} \Big|_{\mathbf{k}=\mathbf{0}} = i E'_\beta \left(-\mathcal{K}_{\text{diff}} \sigma_{\Delta_m}^\alpha t^\beta \right) \left(-\mathcal{K}_{\text{diff}} t^\beta \frac{\partial \sigma_{\Delta_m}^\alpha}{\partial k_j} \right) \Big|_{\mathbf{k}=\mathbf{0}} = 0, \quad (4.7)$$

since $\nabla_{\mathbf{k}} \sigma_{\Delta_m}^\alpha(\mathbf{k})|_{\mathbf{k}=\mathbf{0}} = \mathbf{0}$ by reflection symmetry of the regular simplex.

For the second moment tensor:

$$\langle x_i x_j \rangle(t) = -\frac{\partial^2 \hat{u}(\mathbf{k}, t)}{\partial k_i \partial k_j} \Big|_{\mathbf{k}=\mathbf{0}} = -E'_\beta(0) \left(-\mathcal{K}_{\text{diff}} t^\beta \frac{\partial^2 \sigma_{\Delta_m}^\alpha}{\partial k_i \partial k_j} \right) \Big|_{\mathbf{k}=\mathbf{0}}. \quad (4.8)$$

Since $E_\beta(z) = \frac{1}{\Gamma(1)} + \frac{z}{\Gamma(\beta+1)} + \mathcal{O}(z^2)$, we have $E'_\beta(0) = \frac{1}{\Gamma(\beta+1)}$. By Theorem 2.3, the Hessian of the symbol at $\mathbf{k} = \mathbf{0}$ is:

$$\frac{\partial^2 \sigma_{\Delta_m}^\alpha}{\partial k_i \partial k_j} \Big|_{\mathbf{k}=\mathbf{0}} = \frac{1}{m\alpha} (\mathbf{A}_{m-1})_{ij}. \quad (4.9)$$

Substituting this yields:

$$\langle x_i x_j \rangle(t) = \frac{\mathcal{K}_{\text{diff}}}{m\alpha \Gamma(\beta+1)} (\mathbf{A}_{m-1})_{ij} t^\beta, \quad (4.10)$$

completing the proof. $\square$

Remark 4.4 (Analytical Exact Solution vs. Numerical Intractability). Prior to this formulation, anomalous diffusion in anisotropic media with non-Euclidean triangular/tetrahedral channel networks lacked an exact analytical Green's function, relying on approximate finite-difference or finite-element meshes. Theorem 4.3 resolves this challenge by providing an explicit, closed-form Fourier–Mittag-Leffler propagator directly diagonalized by the A_{m-1} Cartan metric.

5 Conclusion

In this paper, we have formulated and analyzed the physical dynamics governed by the Fractional Simplicial Laplacian introduced in [1]. We established the conservation laws and anisotropic solitary wave solutions of the Nonlinear Simplicial Schrödinger Equation, and derived the exact Mittag-Leffler propagator and Cartan permeability tensor for anomalous porous diffusion. Future research will explore split-step Fourier numerical solvers and experimental transport validation in lithographic microfluidic rock networks.

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